MINING DEALER BEST PRACTICE

HMS Bucket Repair Jig

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

1.0 Introduction..1
2.0 Best Practice Description..................................1
3.0 Implementation Steps..1
4.0 Benefits..4
5.0 Resources Required..4
6.0 Supporting Attachments / References...............4
7.0 Related Best Practices......................................4
8.0 Acknowledgements..5

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1.0 Introduction

The objective of this Best Practice is to increase the safety of personnel performing maintenance work on Hydraulic Mining Shovel (HMS) buckets. Due to the nature of the bucket being a clam shell design, there is only hydraulic power, or to a less controllable degree, gravity, to force it open or closed. During maintenance, when the engine is disconnected and there is no hydraulic power available, a means to secure the bucket so that the clam function doesn't inadvertently open/close should be implemented. The HMS Bucket Repair Jig is a fabricated structure that mechanically restrains the bucket in a set open position to allow service staff to safely perform maintenance work.

2.0 Best Practice Description

A support frame / jig has been designed and fabricated for the purpose of mechanically restraining HMS clam shell style buckets in the open position during services. This is to prevent it from inadvertently closing or moving during a service while the engine is turned off, potentially causing serious injury or death to anyone in the vicinity of the bucket. This Best Practice outlines the key features and benefits of the jig, and how it is transported and implemented on site.

3.0 Implementation Steps

3.1 Design

A) <u>As Seen On Site</u>

Design and fabricate a jig as shown in Figure 1. Going a step further, the bucket mount supports could be designed to be adjustable along dimension C (ie. dimension C on both left and right sides could be shortened or lengthened) to accommodate different bucket mounting positions or bucket sizes. This would be especially useful if the site has various sized HMS models as part of their fleet.

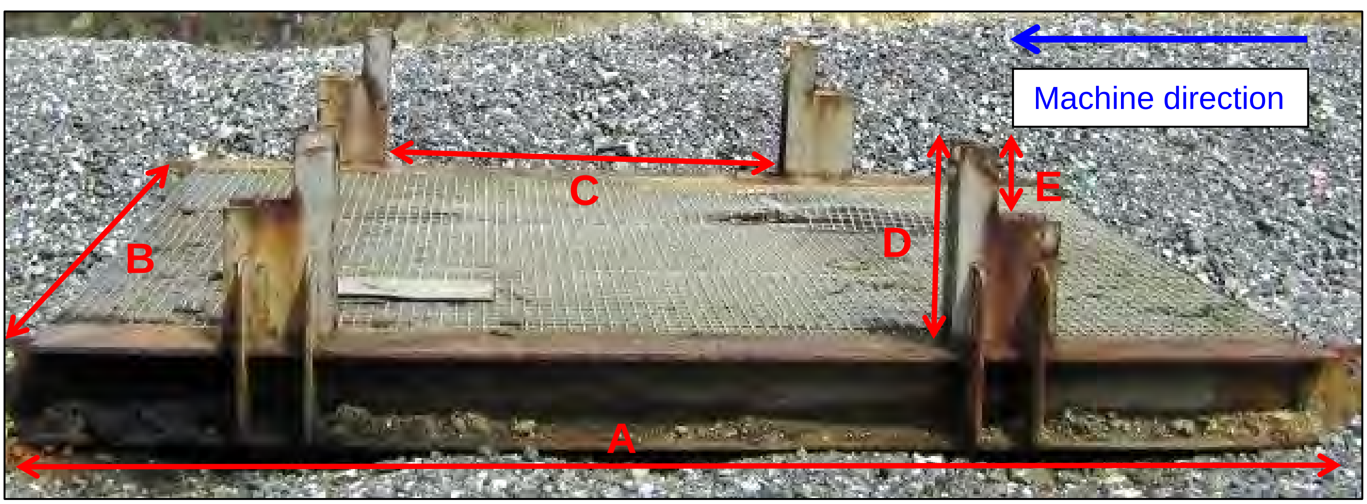


Figure 1: HMS bucket repair jig with example dimensions: A = 440 cm, B = 395 cm, C = 210 cm, D = 72 cm, E = 25 cm.

Figure 2: Close up of one of the four bucket mounts.

Figure 3: HMS bucket secured on the jig.

B) Proposed Additional Design Feature for Improved Bucket Retention

The current design shown above does not secure the bucket in both directions along the bucket edge on the vertical support stands. It provides a physical means via the protruding stopper to retard movement of the bucket in the closing direction, but nothing, apart from friction, to secure the bucket from opening further.

An improved design feature would be to revise the retaining groove to prevent the bucket from opening further as well as closing. A proposed way of doing this is to include a similar stopper on the other side of where the bucket edge rests on the vertical support stands. The proposed design is illustrated in Figure 4.

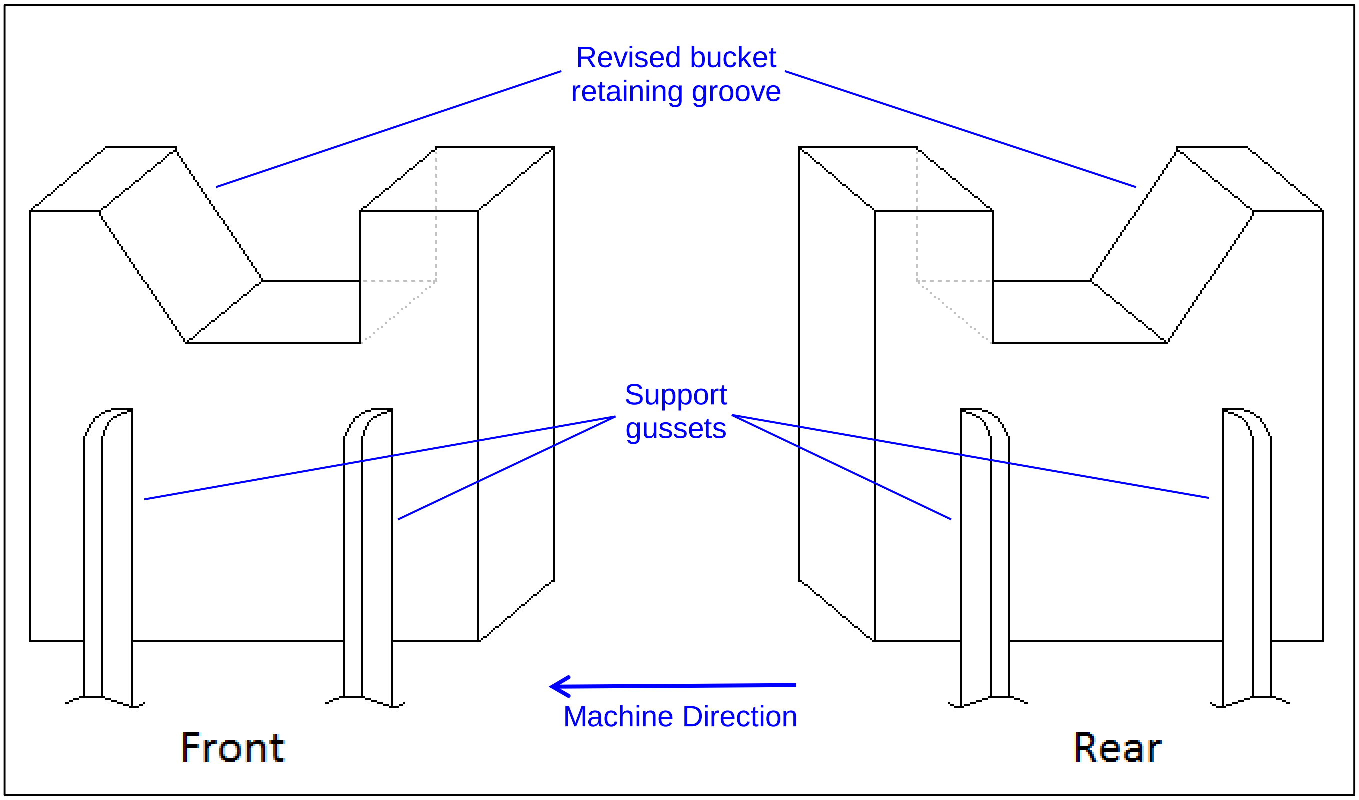
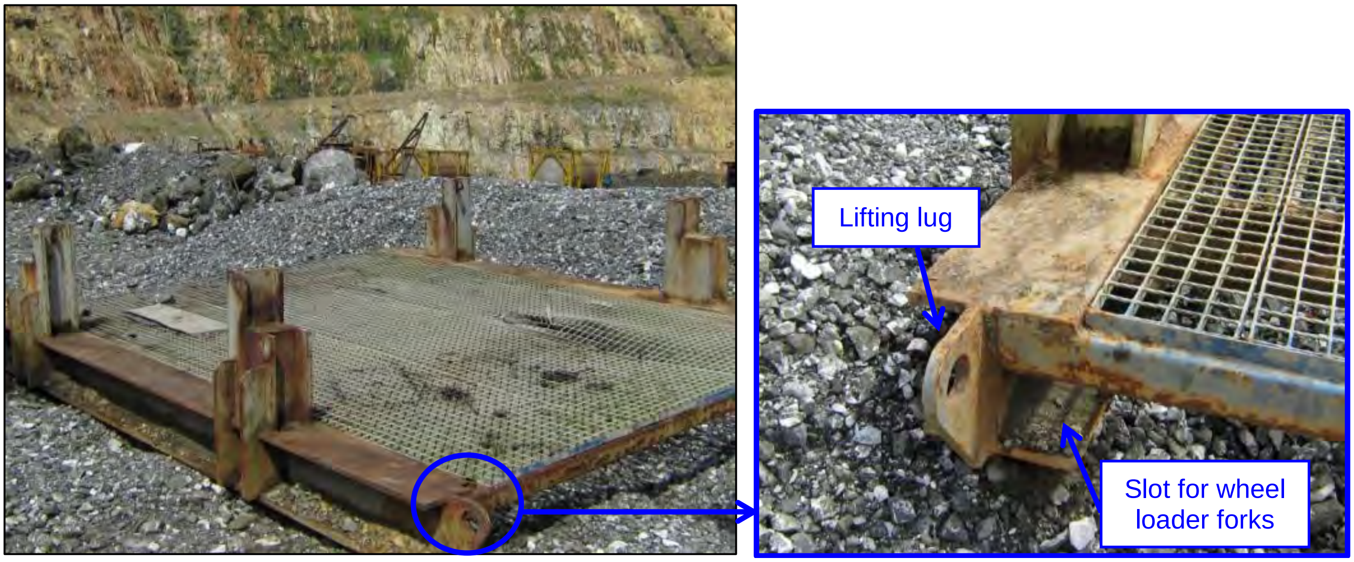


Figure 4: Revised vertical support stand designed to better secure the bucket from any forwards or backwards movement.

3.2 Implementation

When a HMS is down for maintenance or repair work, the HMS bucket repair jig is transported to the machine. This can be done using a wheel loader fitted with a fork attachment to enter the slots that have been cut out. Alternatively, a crane can be used to pick up the jig using the lifting lugs. Refer to Figures 5 & 6 below. Once the jig is in place, the bucket is lowered and positioned in to the four vertical support stands located on the jig, as shown in Figures 2 & 3. Finally, when the bucket is secured safely on the jig and all other maintenance setup procedures have been carried out, the machine can be shut down and tagged out for maintenance work to begin.



Figures 5 & 6: Showing the location of a) the left hand side lifting lug for the crane and b) the left hand side slot for a wheel loader fork, both used for transportation of the jig.

4.0 Benefits

The key benefit of using this jig is safety to maintenance staff. The bucket is mechanically restrained whilst repair or maintenance work is performed, thus eliminating the potential for the bucket to move. An additional benefit is that the platform, as well as support stands which raise the clam shell edges off the ground, allows for easier inspection and access around the bucket.

5.0 Resources Required

- Fabrication of the jig.
- A wheel loader with fork attachment or crane to transport the jig.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgements

This Best Practice was written by:

Adam Rieck
Equipment Management Consultant
Caterpillar Global Mining

Acknowledgement:

Luke Seaton
Product Support Representative – Commercial Engines
Hastings Deering (Aust) Ltd.

Gabriel Aia
Product Support Representative
Hastings Deering (PNG) Ltd.